Business Requirements Document

Banking Portfolio Risk Analytics Dashboard

1. Executive Summary

This project delivers an end-to-end risk analytics framework for a banking portfolio of 3,000 retail, commercial, institutional and private banking clients, with a total loan portfolio of $4.38 billion. The solution combines a SQL-staged database, Python-based data processing and feature engineering, and a four-page Power BI dashboard. Together they give risk officers and relationship managers a single, consistent view of client liquidity and credit exposure, so that financial loss across the lending lifecycle can be minimised.

2. Business Problem

Relationship managers and risk officers do not have a unified view of client liquidity and credit exposure. Deposit, commercial lending and credit line data are held separately, which makes it difficult to assess repayment probability accurately across demographic and income segments.

3. Business Objectives

ID	Objective	Description	Key Metric
BO-1	Credit Exposure Tracking	Consolidate fragmented lending lines into a single view of each client's borrowing.	Total Loan
BO-2	Liquidity Profiling	Aggregate disparate account balances into a single view of each client's liquidity.	Total Deposit
BO-3	Demographic Slicing	Enable drill-downs by Income Band and Nationality to isolate high-risk clusters.	Income Band, Nationality
BO-4	Automated BI Delivery	Provide a production-ready Power BI dashboard for daily operational monitoring.	Power BI dashboard

4. Scope

4.1 In Scope

- Data extraction and schema modelling in SQL.
- Data cleaning, type casting and feature engineering in Python, including Income Band binning.
- Development of four Power BI report pages: Home, Loan Analysis, Deposit Analysis and Summary Dashboard.
- Creation of DAX measures for all core financial KPIs.

4.2 Out of Scope

- Real-time streaming data ingestion.
- Machine learning prediction of live credit scores.

5. Solution Approach

The solution follows a three-stage data pipeline.

Stage	Layer	Description
1	Database Staging (SQL)	Multi-table raw data is ingested into a normalised MySQL relational database to ensure data integrity and schema validation.
2	Data Processing and EDA (Python)	Jupyter notebooks handle missing value audits, datetime parsing, continuous variable binning (Income Bands) and exploratory analysis using pandas, seaborn and matplotlib.
3	Business Intelligence (Power BI)	The curated dataset powers a four-page interactive dashboard with DAX measures for portfolio-level financial KPIs, presented on custom Canva-designed layouts.

6. Deliverables

Deliverable	Description
SQL database	Normalised relational schema holding the staged raw data.
Python data pipeline	Cleaning, type casting and feature engineering, including Income Band binning.
Power BI dashboard	Four pages: Home, Loan Analysis, Deposit Analysis and Summary Dashboard.
DAX measures	Core financial KPIs, including Total Loan and Total Deposit.

7. Baseline Portfolio Insights

The exploratory analysis of the portfolio produced the following findings, which the dashboard is designed to surface and monitor.

- **Exposure concentration by income:** 50.6% of clients are in the Mid income tier ($100k–$300k). This segment holds the highest aggregate debt because of its volume, while High income clients hold the largest per-capita business lending lines.

- **Risk across demographics:** High credit risk (Grades 4 and 5) scales proportionally across European and Asian clients, indicating no localised geographic risk clusters. 68.6% of the portfolio sits in Low Risk Grades 1 and 2.
- **Liquidity consolidation:** A strong positive correlation (0.71) between Bank Deposits and Checking Accounts shows that affluent clients tend to consolidate multiple liquidity facilities within the bank.

Headline Financial Metrics

Metric	Value
Total Loan Exposure	$4.38 billion
Total Deposit Liquidity	$3.77 billion
Projected Annualised Fee Revenue	$170M+

8. Document Control

Item	Detail
Project Title	Banking Portfolio Risk Analytics Dashboard
Document Version	1.0